

Template Title

Paul Haskins
22338905

haskinsp@tcd.ie

Contents

1	Redshift - How do we know the speed of a particle?	3
---	--	---

Throughout this discussion we will work in natural units $c = \hbar = 1$ unless stated otherwise. We are concerned with the Principles of General Relativity. Consider two frames given by the coordinates (t, x, y, z) and (t', x', y', z') . Then we can consider the Lorentz Transformations. In this case we consider a boost in the $\hat{x}$ direction.

$$\begin{aligned}x' &= \gamma (x - \beta t) & \gamma &:= \frac{1}{\sqrt{1 - \beta^2}} \\t' &= \gamma (t - \beta x) \\y' &= y \\z' &= z.\end{aligned}$$

Then the line element is given by

$$\begin{aligned}ds^2 &= -dt^2 + dx^2 + dy^2 + dz^2 \\&= -dt'^2 + dx'^2 + dy'^2 + dz'^2\end{aligned}$$

Where $t > \vec{x}$ is called timelike (massive objects), and $t < \vec{x}$ is spacelike.

$$x^\mu := (t, x, y, z), = (x^0, x^1, x^2, x^3) \quad \mu = 0, 1, 2, 3 \quad (1)$$

The invariant line element is

$$ds^2 = \eta_{\mu\nu} dx^\mu dx^\nu \quad (2)$$

where $\eta_{\mu\nu} = \text{diag} \, (-, +, +, +)$.

What is a general Lorentz transformation which leaves $\eta_{\mu\nu} x^\mu x^\nu$ invariant?

$$\begin{aligned}x'^\mu &= \ell^\mu_\nu x^\nu \\ \eta_{\mu\nu} x'^\mu x'^\nu &= \eta_{\mu\nu} \ell^\mu_\alpha \ell^\nu_\beta x^\alpha x^\beta = \eta_{\alpha\beta} x^\alpha x^\beta.\end{aligned}$$

We can consider quadratic forms given by

$$l^T \eta l = \eta \quad (\det l)^2 = 1 \implies \det l = \pm 1 \quad (3)$$

We will consider Lorentz transformations connected to the positive identity $\mathbb{1}$. If we consider an infinitesimal displacement in spacetime then what is $\Delta t^2 + \Delta x^2 = -\eta_{\mu\nu} \Delta x^\mu \Delta x^\nu$? We can go to the coordinate system tied to this moving object (the primed system).

$$\Delta t'^2 = \Delta t^2 - \Delta x^2 \quad (4)$$

Introduce proper time (the time elapsed between two events as measured by a clock that is stationary relative to the object or observer being tracked.)

$$d\tau = dt \sqrt{1 - \beta^2} \iff \frac{dt}{d\tau} = \gamma \quad (5)$$

$$\beta := \frac{d\vec{x}}{dt} \sim \frac{\Delta \vec{x}}{\Delta t} \quad (6)$$

The clock reads

$$\tau = \int d\tau = \int_0^{t_0} dt \sqrt{1 - \beta^2} \quad (7)$$

The four-velocity is given by $u^\mu = \frac{dx^\mu}{d\tau}$. $A_\mu = \eta_{\mu\nu} A^\nu$.

$$\begin{aligned} u_\mu u^\mu &= \eta_{\mu\nu} u^\mu u^\nu = \eta_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = - \left(\frac{dt}{d\tau} \right)^2 + \left(\frac{d\vec{x}}{d\tau} \right)^2 \\ &= \left(\frac{dt}{d\tau} \right)^2 \left[-1 + \left(\frac{d\vec{x}}{d\tau} \right) \left(\frac{d\tau}{dt} \right)^2 \right]. \end{aligned}$$

Thus we have

1 Redshift - How do we know the speed of a particle?

Timelike geodesic

$$t^\mu = \frac{dx^\mu}{d\tau} = \frac{p^\mu}{m}$$

$$t^\mu t_\mu = 1.$$

Take motion in $\hat{z}$ direction $p^\mu = [\gamma \ 0 \ 0 \ \gamma\beta]$ we can infer the speed β of the particle from

$$\gamma = \frac{1}{\sqrt{1-\beta^2}}, \quad \gamma = -u^\mu p_\mu = -u \cdot p \quad (1.1)$$

What about p^μ ?

$$t^\mu = \frac{dx^\mu}{d\lambda} \quad (1.2)$$

We can't set $\lambda = \tau$ since $d\tau = 0$, that we can take any linear combination of $\lambda = a\lambda + b$. We can demand in a locally inertial frame the tangent is the 4 momentum.

$$t^\mu = p^\mu = [\varepsilon \ 0 \ 0 \ \varepsilon] \quad (1.3)$$

An observer at rest in this inertial frame has

$$u^{\hat{\mu}} = [1 \ 0 \ 0 \ 0] \implies u_{\hat{\mu}} = [-1 \ 0 \ 0 \ 0] \quad (1.4)$$

So

$$p^{\hat{\mu}} u_{\hat{\mu}} = -\varepsilon \quad (1.5)$$

Consider a static observer ($u \sim k = \frac{\partial}{\partial t}$) the schwartzschild metric has a killing vector. since the metric doesn't depend on t thus $K^\mu = (\partial_t)^\mu$ is a killing vector. $u^\mu \propto K^\mu$, we can't just set $u^\mu = K^\mu$ as the 4-velocity must obey $u_\mu u^\mu = -1$ for timelike geodesics. Suppose

$$u^\mu = AK^\mu \implies u^\mu u_\mu = A^2 K^\mu K_\mu = A^2 K^2 \quad (1.6)$$

Demand $A^2 K^2 = -1$

$$A = \frac{1}{\sqrt{-K^2}} \quad (1.7)$$

Thus

$$u^\mu = \frac{K^\mu}{\sqrt{-K^2}} \quad (1.8)$$

where we can find K^2 for a given killing vector using

$$K^2 = g_{\mu\nu} K^\mu K^\nu \quad (1.9)$$

for the Schwartzschild and RN solution this is just the first component but things are more complicated for the Kerr case.

Recall for a geodesic paramterised by k ,

$$\nabla_t (t \cdot k) = 0 \quad (1.10)$$

where t is the tangent to the photon's geodesic.

$$\frac{\omega_1}{\omega_2} = \frac{\varepsilon_1}{\varepsilon_2} = \frac{-p \cdot k_1}{-p \cdot k_2} = \frac{p \cdot k_1}{\sqrt{-K_1^2}} \cdot \frac{\sqrt{-K_2^2}}{p \cdot k_2} = \sqrt{\frac{K_2^2}{K_1^2}}.$$

For schwatzschild $k = \frac{\partial}{\partial t}$

$$K^2 = +g_{00} = -\left(1 - \frac{2GM}{r}\right) \tag{1.11}$$

$$\frac{\omega_1}{\omega_2} = \frac{\left(1 - \frac{2GM}{r_2}\right)^{\frac{1}{2}}}{\left(1 - 2\frac{GM}{r_1}\right)^{\frac{1}{2}}} \tag{1.12}$$

kkkkkkkkkkkkkkkkkk